

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE:CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I S. Adhi kesavan / 192521172(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Adhi Kesavan

APPLICATION- BASED QUESTIONS

1 . Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

2 . Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Solution :

1. Scenario: Suspicious Digital Asset Investigation

a. Analyze how investigators could detect the presence of hidden data within the image.

Investigators can detect hidden data (steganography) using the following methods:

1. Visual Inspection

- Compare the suspicious image with the original image.
- Look for unusual colors , noise, or pixel distortions.

2. Statistical Analysis

- Analyze pixel values and color distributions.
- Detect abnormal patterns caused by hidden data.

3. Histogram Analysis

- Examine the image histogram for irregularities.
- Hidden data often changes the histogram distribution.

4. Least Significant Bit (LSB) Analysis

- Check whether the least significant bits of image pixels have been modified.
- Many steganography techniques hide data in these bits.

5. Steganalysis Tools

- Use forensic software such as:
 - StegExpose
 - StegDetect

- OpenStego
- Forensically
- These tools automatically detect hidden information.

b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Technique 1: Encrypt the Secret Data Before Hiding

- Encrypt the confidential information using algorithms like **AES** before embedding it.
- Even if the hidden data is extracted, it cannot be understood without the encryption key.

Technique 2: Use Randomized or Adaptive Steganography

- Instead of embedding data sequentially, hide it in randomly selected pixels using a secret key.
- This makes detection through statistical analysis much more difficult.

2 . Scenario: Legacy Encryption in Corporate Communication

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

a. Determine the most probable original message by reversing the encryption process.

This ciphertext is produced using a **Rail Fence Cipher (3 rails)**.

Original Plaintext:

WE ARE DISCOVERED FLEE AT ONCE

(Removing spaces:)

WEAREDISCOVEREDFLEEATONCE

b. Critically evaluate why such a method is vulnerable in modern communication systems.

The Rail Fence (transposition) cipher has several weaknesses:

1. **No Substitution**
 - The letters remain unchanged; only their positions are rearranged.
2. **Easy Brute Force**
 - Attackers can try different numbers of rails until the message becomes readable.
3. **Pattern Preservation**
 - Letter frequencies remain the same, making cryptanalysis easier.
4. **No Secret Key Complexity**
 - The key space is very small (number of rails), making attacks simple.
5. **Not Secure Against Modern Computers**
 - Modern computers can break such ciphers within seconds.

c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Use AES (Advanced Encryption Standard).

Advantages:

- Strong symmetric encryption algorithm.
- Supports 128-bit, 192-bit, and 256-bit keys.
- Resistant to brute-force and cryptanalysis attacks.
- Widely used in banking, government, military, and secure internet communications.

Alternatively, the organization could first encrypt the message using AES and then, if desired, apply a transposition cipher as an additional obfuscation layer.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation